

Second Report: Results of Numerical Methods

Numerical Analysis

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LU Factorization with Simple Gaussian Elimination

Stage 0

$$M = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 1.000000 & 15.500000 & 3.000000 & 8.000000 \\ 0.000000 & -1.300000 & -4.000000 & 1.100000 \\ 14.000000 & 5.000000 & -2.000000 & 30.000000 \end{pmatrix}$$
$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad U = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Stage 1

$$M = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & -1.300000 & -4.000000 & 1.100000 \\ 0.000000 & 8.500000 & -2.000000 & 19.500000 \end{pmatrix}$$
$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.250000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 3.500000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

Stage 2

$$M = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & 0.000000 & -3.752381 & 1.698413 \\ 0.000000 & 0.000000 & -3.619048 & 15.587302 \end{pmatrix}$$
$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.250000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & -0.082540 & 1.000000 & 0.000000 \\ 3.500000 & 0.539683 & 0.000000 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & 0.000000 & -3.752381 & 1.698413 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

Stage 3

$$M = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & 0.000000 & -3.752381 & 1.698413 \\ 0.000000 & 0.000000 & 0.000000 & 13.949239 \end{pmatrix}$$

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.250000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & -0.082540 & 1.000000 & 0.000000 \\ 3.500000 & 0.539683 & 0.964467 & 1.000000 \end{pmatrix}$$

$$U = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & 0.000000 & -3.752381 & 1.698413 \\ 0.000000 & 0.000000 & 0.000000 & 13.949239 \end{pmatrix}$$

Solution x

$$\begin{aligned} x[0] &= 0.5251091703 & x[1] &= 0.2554585153 \\ x[2] &= -0.4104803493 & x[3] &= -0.2816593886 \end{aligned}$$

Evidence

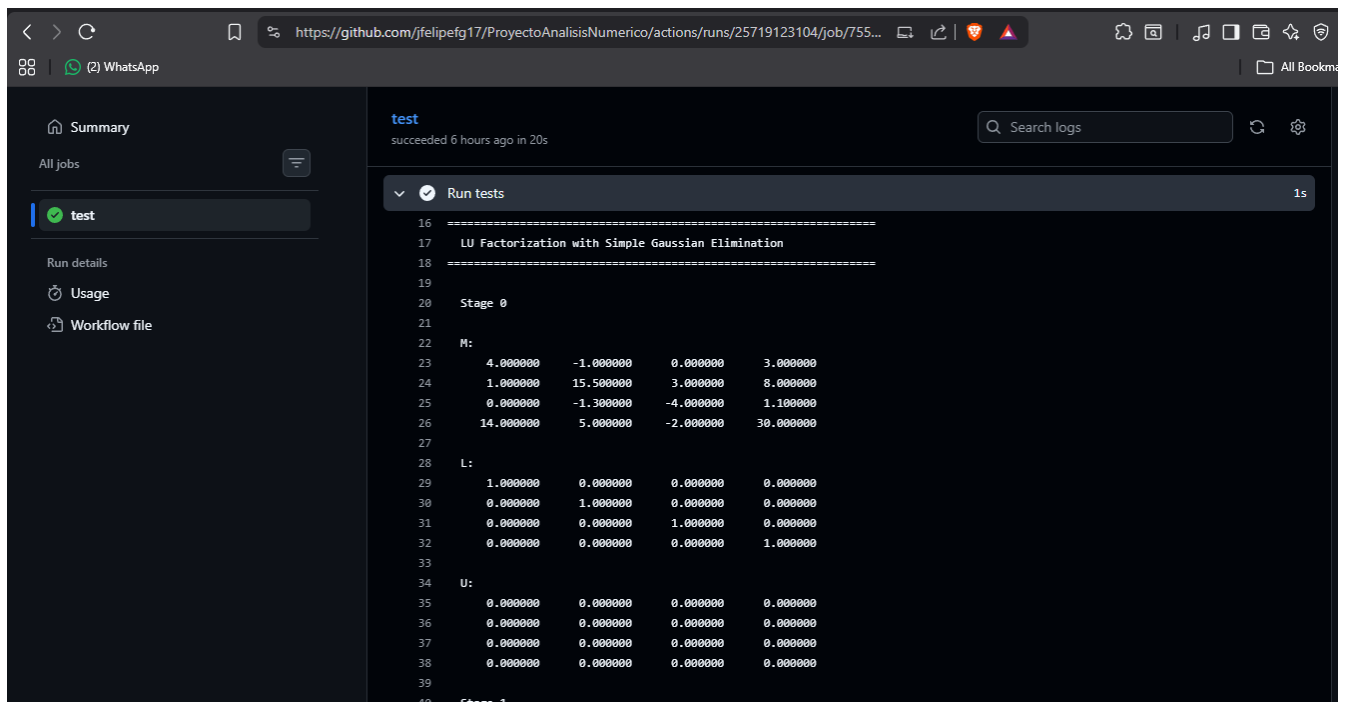


Figure 1: *
GitHub execution screenshot

LU Factorization with Partial Pivoting

Stage 0

$$M = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 1.000000 & 15.500000 & 3.000000 & 8.000000 \\ 0.000000 & -1.300000 & -4.000000 & 1.100000 \\ 14.000000 & 5.000000 & -2.000000 & 30.000000 \end{pmatrix}$$

$$P = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

$$U = \begin{pmatrix} 0.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

Stage 1

$$M = \begin{pmatrix} 14.000000 & 5.000000 & -2.000000 & 30.000000 \\ 0.000000 & 15.142857 & 3.142857 & 5.857143 \\ 0.000000 & -1.300000 & -4.000000 & 1.100000 \\ 0.000000 & -2.428571 & 0.571429 & -5.571429 \end{pmatrix}$$

$$P = \begin{pmatrix} 0.000000 & 0.000000 & 0.000000 & 1.000000 \\ 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 1.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.071429 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.285714 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

$$U = \begin{pmatrix} 14.000000 & 5.000000 & -2.000000 & 30.000000 \\ 0.000000 & 15.142857 & 3.142857 & 5.857143 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

Stage 2

$$M = \begin{pmatrix} 14.000000 & 5.000000 & -2.000000 & 30.000000 \\ 0.000000 & 15.142857 & 3.142857 & 5.857143 \\ 0.000000 & 0.000000 & -3.730189 & 1.602830 \\ 0.000000 & 0.000000 & 1.075472 & -4.632075 \end{pmatrix}$$

$$P = \begin{pmatrix} 0.000000 & 0.000000 & 0.000000 & 1.000000 \\ 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 1.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.071429 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & -0.085849 & 1.000000 & 0.000000 \\ 0.285714 & -0.160377 & 0.000000 & 1.000000 \end{pmatrix}$$

$$U = \begin{pmatrix} 14.000000 & 5.000000 & -2.000000 & 30.000000 \\ 0.000000 & 15.142857 & 3.142857 & 5.857143 \\ 0.000000 & 0.000000 & -3.730189 & 1.602830 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

Stage 3

$$M = \begin{pmatrix} 14.000000 & 5.000000 & -2.000000 & 30.000000 \\ 0.000000 & 15.142857 & 3.142857 & 5.857143 \\ 0.000000 & 0.000000 & -3.730189 & 1.602830 \\ 0.000000 & 0.000000 & 0.000000 & -4.169954 \end{pmatrix}$$

$$P = \begin{pmatrix} 0.000000 & 0.000000 & 0.000000 & 1.000000 \\ 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 1.000000 & 0.000000 & 0.000000 & 0.000000 \end{pmatrix}$$

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.071429 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & -0.085849 & 1.000000 & 0.000000 \\ 0.285714 & -0.160377 & -0.288316 & 1.000000 \end{pmatrix}$$

$$U = \begin{pmatrix} 14.000000 & 5.000000 & -2.000000 & 30.000000 \\ 0.000000 & 15.142857 & 3.142857 & 5.857143 \\ 0.000000 & 0.000000 & -3.730189 & 1.602830 \\ 0.000000 & 0.000000 & 0.000000 & -4.169954 \end{pmatrix}$$

Solution x

$$\begin{aligned} x[0] &= 0.5251091703 & x[1] &= 0.2554585153 \\ x[2] &= -0.4104803493 & x[3] &= -0.2816593886 \end{aligned}$$

Evidence

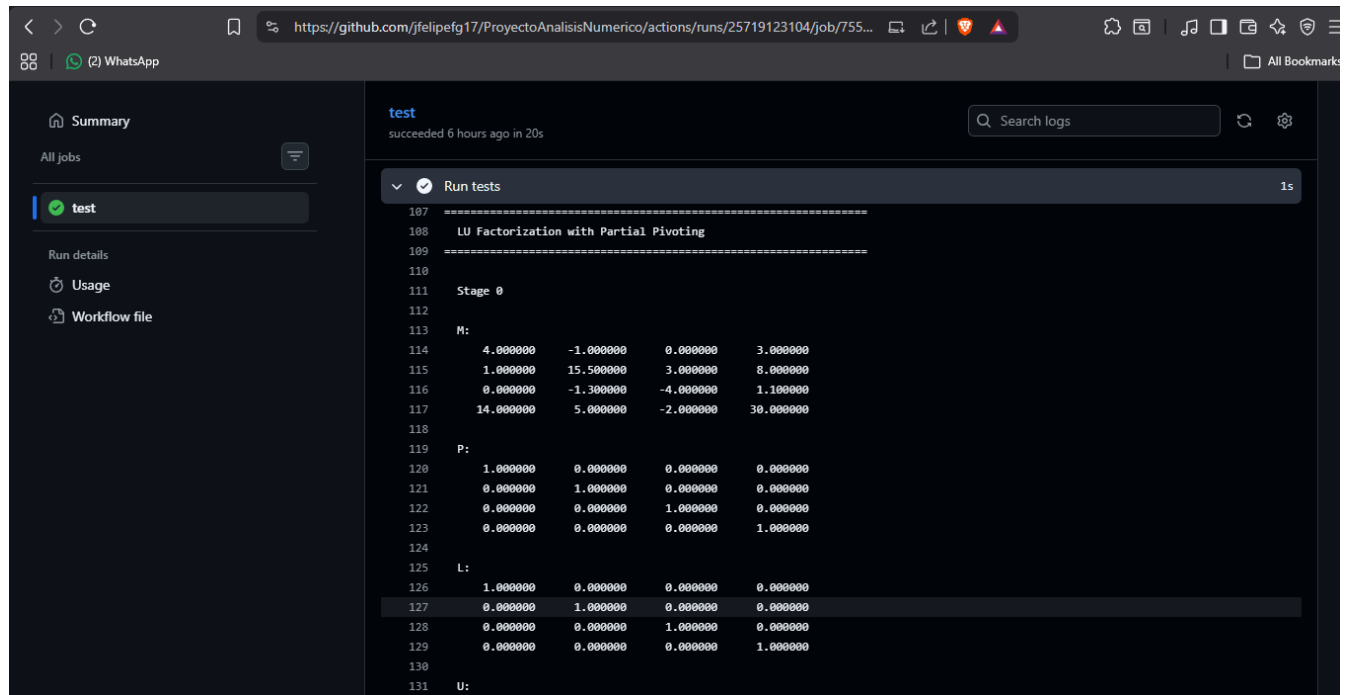


Figure 2: *
GitHub execution screenshot

Crout Factorization

Stage 0

$$A = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 1.000000 & 15.500000 & 3.000000 & 8.000000 \\ 0.000000 & -1.300000 & -4.000000 & 1.100000 \\ 14.000000 & 5.000000 & -2.000000 & 30.000000 \end{pmatrix}$$

Stage 1

$$L = \begin{pmatrix} 4.000000 & 0.000000 & 0.000000 & 0.000000 \\ 1.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 14.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 1.000000 & -0.250000 & 0.000000 & 0.750000 \\ 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Stage 2

$$L = \begin{pmatrix} 4.000000 & 0.000000 & 0.000000 & 0.000000 \\ 1.000000 & 15.750000 & 0.000000 & 0.000000 \\ 0.000000 & -1.300000 & 1.000000 & 0.000000 \\ 14.000000 & 8.500000 & 0.000000 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 1.000000 & -0.250000 & 0.000000 & 0.750000 \\ 0.000000 & 1.000000 & 0.190476 & 0.460317 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Stage 3

$$L = \begin{pmatrix} 4.000000 & 0.000000 & 0.000000 & 0.000000 \\ 1.000000 & 15.750000 & 0.000000 & 0.000000 \\ 0.000000 & -1.300000 & -3.752381 & 0.000000 \\ 14.000000 & 8.500000 & -3.619048 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 1.000000 & -0.250000 & 0.000000 & 0.750000 \\ 0.000000 & 1.000000 & 0.190476 & 0.460317 \\ 0.000000 & 0.000000 & 1.000000 & -0.452623 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Solution x

$$\begin{aligned} x[0] &= 0.5251091703 & x[1] &= 0.2554585153 \\ x[2] &= -0.4104803493 & x[3] &= -0.2816593886 \end{aligned}$$

Evidence

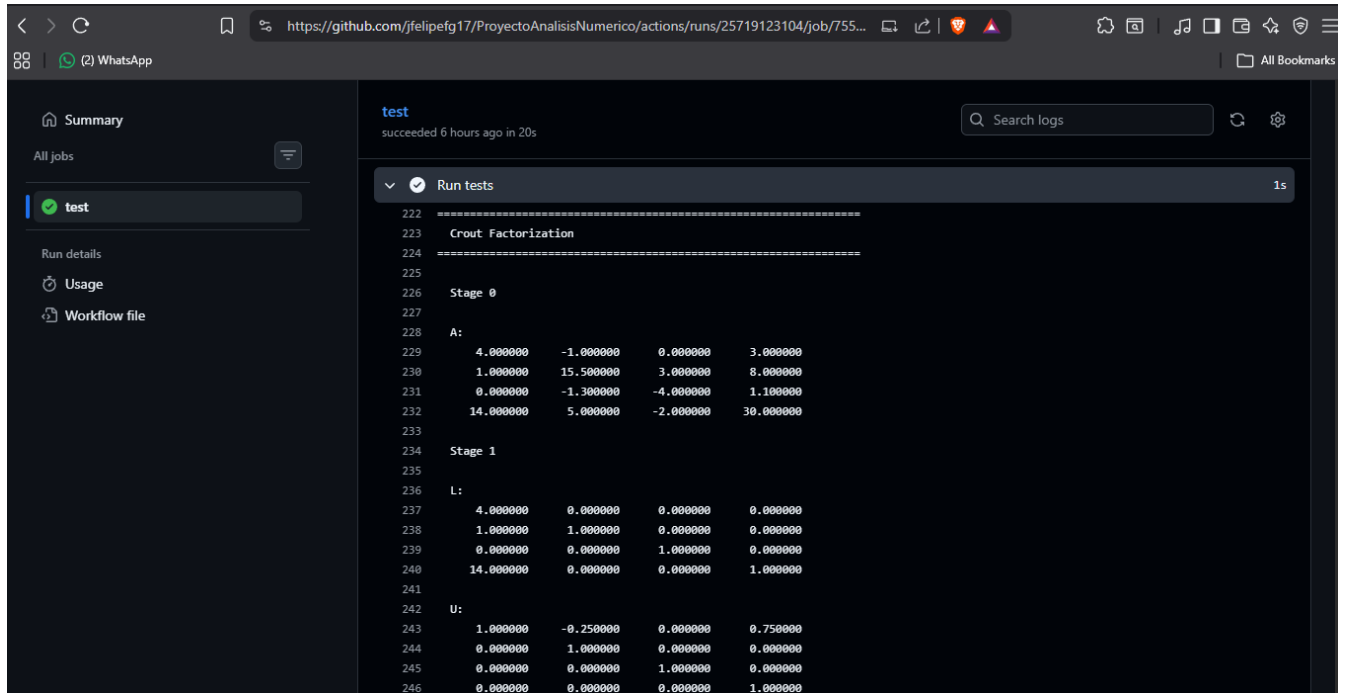


Figure 3: *
GitHub execution screenshot

Doolittle Factorization

Stage 0

$$A = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 1.000000 & 15.500000 & 3.000000 & 8.000000 \\ 0.000000 & -1.300000 & -4.000000 & 1.100000 \\ 14.000000 & 5.000000 & -2.000000 & 30.000000 \end{pmatrix}$$

Stage 1

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.250000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 3.500000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Stage 2

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.250000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & -0.082540 & 1.000000 & 0.000000 \\ 3.500000 & 0.539683 & 0.000000 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Stage 3

$$L = \begin{pmatrix} 1.000000 & 0.000000 & 0.000000 & 0.000000 \\ 0.250000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & -0.082540 & 1.000000 & 0.000000 \\ 3.500000 & 0.539683 & 0.964467 & 1.000000 \end{pmatrix}$$
$$U = \begin{pmatrix} 4.000000 & -1.000000 & 0.000000 & 3.000000 \\ 0.000000 & 15.750000 & 3.000000 & 7.250000 \\ 0.000000 & 0.000000 & -3.752381 & 1.698413 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Solution x

$$\begin{aligned} x[0] &= 0.5251091703 & x[1] &= 0.2554585153 \\ x[2] &= -0.4104803493 & x[3] &= -0.2816593886 \end{aligned}$$

Evidence

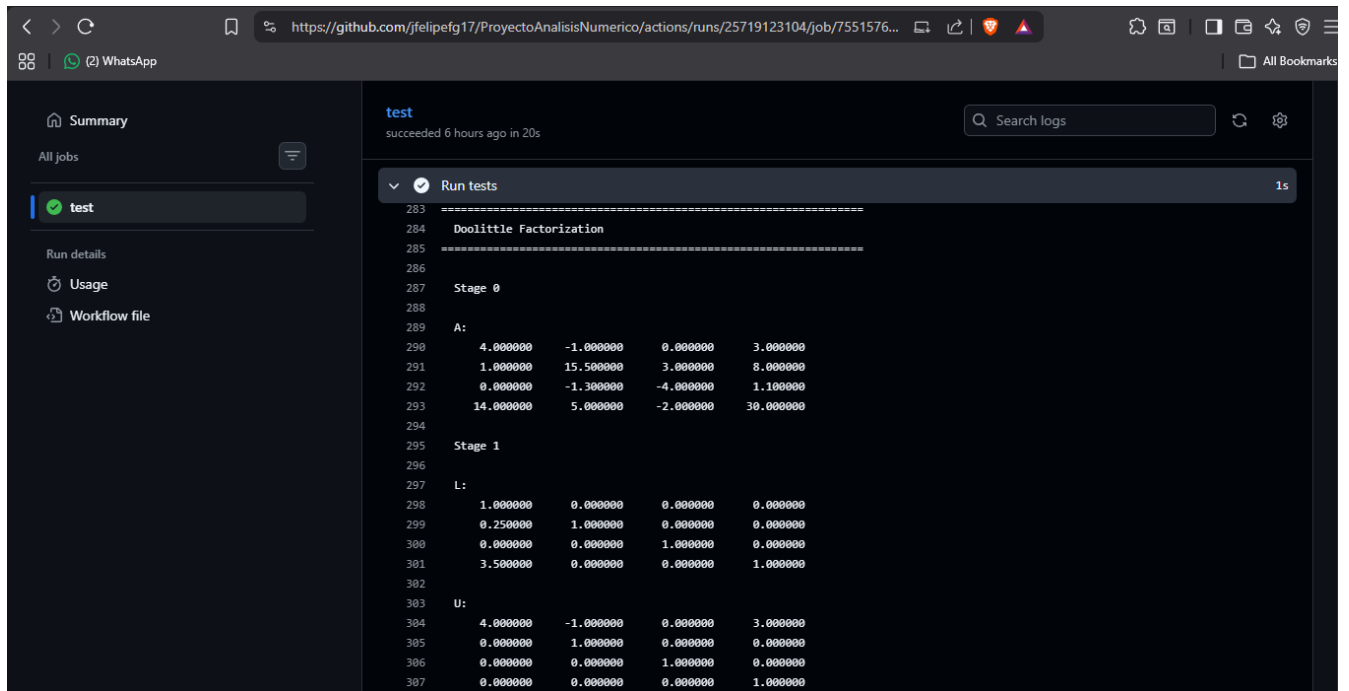


Figure 4: *
GitHub execution screenshot

Cholesky Factorization

Input

$$A = \begin{pmatrix} 4.000000 & 2.000000 & 2.000000 \\ 2.000000 & 5.000000 & 3.000000 \\ 2.000000 & 3.000000 & 6.000000 \end{pmatrix} \quad b = \begin{pmatrix} 8 \\ 10 \\ 11 \end{pmatrix}$$

Stage 0

$$A = \begin{pmatrix} 4.000000 & 2.000000 & 2.000000 \\ 2.000000 & 5.000000 & 3.000000 \\ 2.000000 & 3.000000 & 6.000000 \end{pmatrix}$$
$$L = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad U = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Stage 1

$$L = \begin{pmatrix} 2.000000 & 0.000000 & 0.000000 \\ 1.000000 & 1.000000 & 0.000000 \\ 1.000000 & 0.000000 & 1.000000 \end{pmatrix} \quad U = \begin{pmatrix} 2.000000 & 1.000000 & 1.000000 \\ 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Stage 2

$$L = \begin{pmatrix} 2.000000 & 0.000000 & 0.000000 \\ 1.000000 & 2.000000 & 0.000000 \\ 1.000000 & 1.000000 & 1.000000 \end{pmatrix} \quad U = \begin{pmatrix} 2.000000 & 1.000000 & 1.000000 \\ 0.000000 & 2.000000 & 1.000000 \\ 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Final Factorization $A = L \cdot L^T$

$$L = \begin{pmatrix} 2.000000 & 0.000000 & 0.000000 \\ 1.000000 & 2.000000 & 0.000000 \\ 1.000000 & 1.000000 & 2.000000 \end{pmatrix} \quad U = L^T = \begin{pmatrix} 2.000000 & 1.000000 & 1.000000 \\ 0.000000 & 2.000000 & 1.000000 \\ 0.000000 & 0.000000 & 2.000000 \end{pmatrix}$$

Solution x

$$x[0] = 1.00000000000 \quad x[1] = 1.00000000000 \quad x[2] = 1.00000000000$$

Evidence

```
=====
Cholesky Factorization | A = L * L^T
=====

Stage 0
A:
  4.000000    2.000000    2.000000
  2.000000    5.000000    3.000000
  2.000000    3.000000    6.000000

Stage 1
L:
  2.000000    0.000000    0.000000
  1.000000    1.000000    0.000000
  1.000000    0.000000    1.000000

U:
  2.000000    1.000000    1.000000
  0.000000    1.000000    0.000000
  0.000000    0.000000    1.000000

Stage 2
L:
  2.000000    0.000000    0.000000
  1.000000    2.000000    0.000000
  1.000000    1.000000    1.000000

U:
  2.000000    1.000000    1.000000
  0.000000    2.000000    1.000000
  0.000000    0.000000    1.000000

Solution x:
x[0] = 1.0000000000
x[1] = 1.0000000000
x[2] = 1.0000000000
Verification L @ U == A : True
```

Figure 5: Enter Caption

Figure 6: *
GitHub execution screenshot

Vandermonde Interpolation

Test 1 — Three points ($y = x^2$)

Input data

i	0	1	2
x_i	1.0	2.0	3.0
y_i	1.0	4.0	9.0

Vandermonde matrix

$$A = \begin{pmatrix} 1.000000 & 1.000000 & 1.000000 \\ 4.000000 & 2.000000 & 1.000000 \\ 9.000000 & 3.000000 & 1.000000 \end{pmatrix}$$

Polynomial coefficients (descending order)

$$\text{Coef} = (1.000000 \quad 0.000000 \quad 0.000000)$$

Interpolating polynomial

$$p(x) = 1.000000 \cdot x^2 + 0.000000 \cdot x + 0.000000$$

Evaluation table

i	x_i	y_i	$p(x_i)$	error
0	1.0000000000	1.0000000000	1.0000000000	0.0000×10^0
1	2.0000000000	4.0000000000	4.0000000000	0.0000×10^0
2	3.0000000000	9.0000000000	9.0000000000	0.0000×10^0

Test 2 — Four points

Input data

i	0	1	2	3
x_i	0.0	1.0	2.0	3.0
y_i	1.0	2.0	5.0	10.0

Vandermonde matrix

$$A = \begin{pmatrix} 0.000000 & 0.000000 & 0.000000 & 1.000000 \\ 1.000000 & 1.000000 & 1.000000 & 1.000000 \\ 8.000000 & 4.000000 & 2.000000 & 1.000000 \\ 27.000000 & 9.000000 & 3.000000 & 1.000000 \end{pmatrix}$$

Polynomial coefficients (descending order)

$$\text{Coef} = (0.000000 \quad 1.000000 \quad 0.000000 \quad 1.000000)$$

Interpolating polynomial

$$p(x) = 0.000000 \cdot x^3 + 1.000000 \cdot x^2 + 0.000000 \cdot x + 1.000000$$

Evaluation table

i	x_i	y_i	$p(x_i)$	error
0	0.0000000000	1.0000000000	1.0000000000	0.0000×10^0
1	1.0000000000	2.0000000000	2.0000000000	0.0000×10^0
2	2.0000000000	5.0000000000	5.0000000000	0.0000×10^0
3	3.0000000000	10.0000000000	10.0000000000	0.0000×10^0

Evidence

```
=====
Vandermonde | 3 points | y = x^2 expected
=====
p(x) = 1.000000*x^2 + 0.000000*x + 0.000000
i      xi      yi      p(xi)      error
-----
0      1.0000000000  1.0000000000  1.0000000000  0.0000e+00
1      2.0000000000  4.0000000000  4.0000000000  0.0000e+00
2      3.0000000000  9.0000000000  9.0000000000  0.0000e+00
=====
Vandermonde | 4 points
=====
p(x) = 0.000000*x^3 + 1.000000*x^2 + 0.000000*x + 1.000000
i      xi      yi      p(xi)      error
-----
0      0.0000000000  1.0000000000  1.0000000000  0.0000e+00
1      1.0000000000  2.0000000000  2.0000000000  0.0000e+00
2      2.0000000000  5.0000000000  5.0000000000  0.0000e+00
3      3.0000000000  10.0000000000 10.0000000000  0.0000e+00
=====
```

Figure 7: Enter Caption

Figure 8: *
GitHub execution screenshot

Newton Interpolation

Input data

i	0	1	2	3
x_i	-1	0	3	4
y_i	15.5	3	8	1

Divided Differences Table

15.500000	-12.500000	3.5416667	-1.1416667
3.000000	1.6666667	-2.1666667	0.0000000
8.000000	-7.0000000	0.0000000	0.0000000
1.000000	0.0000000	0.0000000	0.0000000

Newton coefficients

$$\text{Coef} = (15.500000 \quad -12.500000 \quad 3.5416667 \quad -1.1416667)$$

Newton polynomial

$$p(x) = -1.14166666666667x^3 + 5.825x^2 - 5.53333333333333x + 3.0$$

Console output

Divided Differences Table:

```
[[ 15.5      -12.5      3.5416667    -1.1416667]
 [  3.       1.6666667   -2.1666667     0.         ]
 [  8.       -7.        0.          0.         ]
 [  1.        0.         0.          0.         ]]
```

Newton coefficients:

```
[ 15.5      -12.5      3.5416667    -1.1416667]
```

Newton polynomial:

```
-1.14166666666667*x**3 + 5.825*x**2 - 5.53333333333333*x + 3.0
```

Lagrange Interpolation

Input data

i	0	1	2	3
x_i	-1	0	3	4
y_i	15.5	3	8	1

Lagrange basis polynomials

$$L_0(x) = -\frac{x^3}{20} + \frac{7x^2}{20} - \frac{3x}{5}$$

$$L_1(x) = \frac{x^3}{12} - \frac{x^2}{2} + \frac{5x}{12} + 1$$

$$L_2(x) = -\frac{x^3}{12} + \frac{x^2}{4} + \frac{x}{3}$$

$$L_3(x) = \frac{x^3}{20} - \frac{x^2}{10} - \frac{3x}{20}$$

Lagrange polynomial

$$p(x) = -1.14166666666667x^3 + 5.825x^2 - 5.53333333333333x + 3$$

Polynomial coefficients

$$\text{Coef} = (-1.14166666666667 \quad 5.82500000000000 \quad -5.53333333333333 \quad 3.00000000000000)$$

Console output

Lagrange basis polynomials:

$$L_0(x) = -x^{**3}/20 + 7*x^{**2}/20 - 3*x/5$$

$$L_1(x) = x^{**3}/12 - x^{**2}/2 + 5*x/12 + 1$$

$$L_2(x) = -x^{**3}/12 + x^{**2}/4 + x/3$$

$$L_3(x) = x^{**3}/20 - x^{**2}/10 - 3*x/20$$

Lagrange polynomial:

$$-1.14166666666667*x^{**3} + 5.825*x^{**2} - 5.53333333333333*x + 3$$

Polynomial coefficients:

$$[-1.14166666666667, 5.82500000000000, -5.53333333333333, 3.00000000000000]$$